



PLANET 3

TRAPPIST-1 E

- Description:** Temperate rocky planet orbiting in the habitable zone, where liquid water could exist. It is probably tidally locked.
- Host star:** TRAPPIST-1, an ultracool red dwarf barely larger than Jupiter and only about 9% the mass of the Sun.
- System:** One of 7 known planets orbiting TRAPPIST-1, and the fourth one out from the star.
- Location:** Constellation Aquarius, about 41 light-years from Earth. Far too faint to see without a telescope (magnitude 17).
- Discovery:** In 2017 by Michaël Gillon et al., using the TRAPPIST telescope in Chile together with NASA's Spitzer Space Telescope.

Datasheet

EXOPLANET DATA		
	TEMPERATURE	-24 °C
	PRESSURE	Unknown
	ATMOSPHERE	Unknown
	GRAVITY	0.82 g
	PLANET RADIUS	
	ORBITAL PERIOD	
	SEMI-MAJOR AXIS	
	TRANSIT DEPTH	
	TRANSIT DURATION	

HOST STAR DATA		
	NAME	TRAPPIST-1
	SPECTRAL TYPE	M8V
	TEMPERATURE	2566 K
	DISTANCE	41 ly
	MASS	0.090 M _☉
	STAR RADIUS	0.119 R _☉
	AGE	7.6 Gyr
	APPARENT MAGNITUDE	17.02
	NUMBER OF PLANETS	7

SURVIVAL CHALLENGE

Intense radiation from an active red dwarf star

NOTES



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NOTES

PHYSICS

 PLANET RADIUS	
 AI-ASSISTANCE	
 REFLECTIONS	

MATHEMATICS

 ORBITAL PERIOD		 SEMI-MAJOR AXIS	
 AI-ASSISTANCE			
 REFLECTIONS			

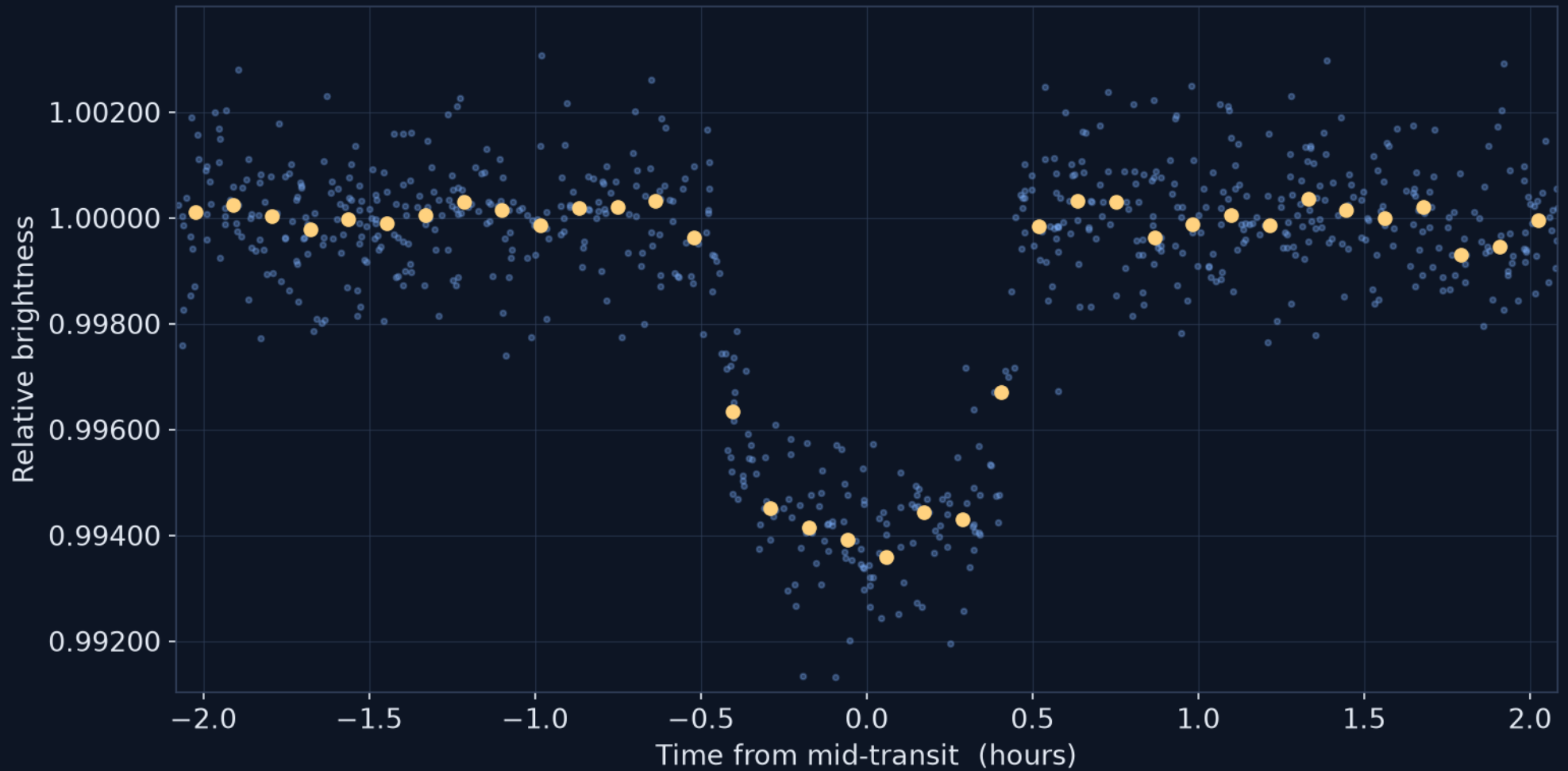
CHEMISTRY

 SAMPLE DATA	
 AI-ASSISTANCE	
 REFLECTIONS	

BIOLOGY

 ORGANISM	
 AI-ASSISTANCE	
 REFLECTIONS	

TRAPPIST-1 e: single transit



Blue points are simulated brightness measurements (many orbits phase-folded together, like real data); the gold points are binned averages.

TRAPPIST-1 e: light curve over several orbits

